STUDENT ATTENDANCE MONITORING SYSTEM

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Adviser

GROUP 1

CPE21S1 - CPE 010
Section - Course Code



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PROJECT OBJECTIVE

GENERAL OBJECTIVE

- To develop a C++ Student Attendance Monitoring System that emulates LMS attendance workflows using customized data structures and algorithms

SPECIFIC OBJECTIVE

- To Manage student attendance using First-In, First-Out (FIFO) queue
- To search student names and attendance records efficiently using binary search
- To sort student names and attendance records using merge sort.
- To analyze the time complexity of the main operations theoretically and experimentally.

Scope and delimitations

SCOPE

- Records student attendance categorized by date and time.
- Generates randomized or patterned (round-robin) student groupings.
- Looks up individual student records and attendance history.
- Target Users: Instructors managing class attendance and student participation.
- Data Storage: Saves student information and logs locally in a master CSV file.

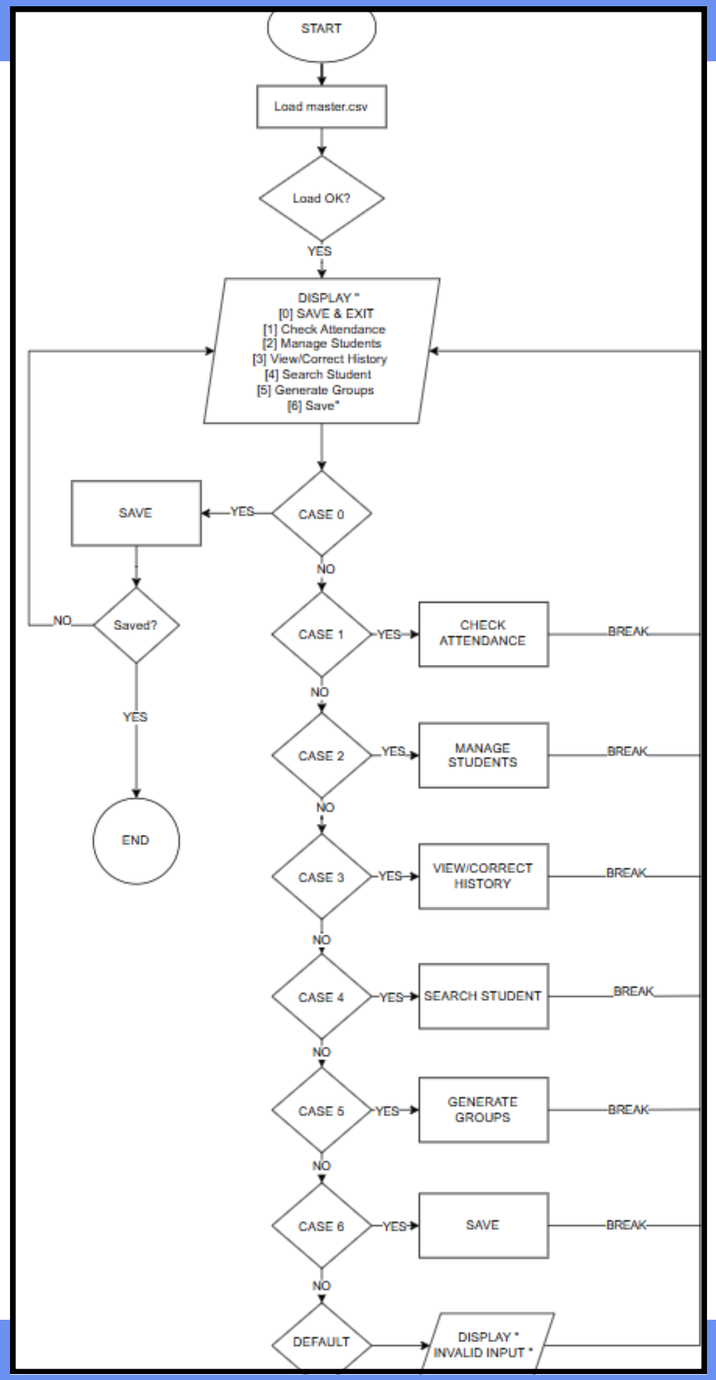
Scope and delimitations

DELIMITATIONS

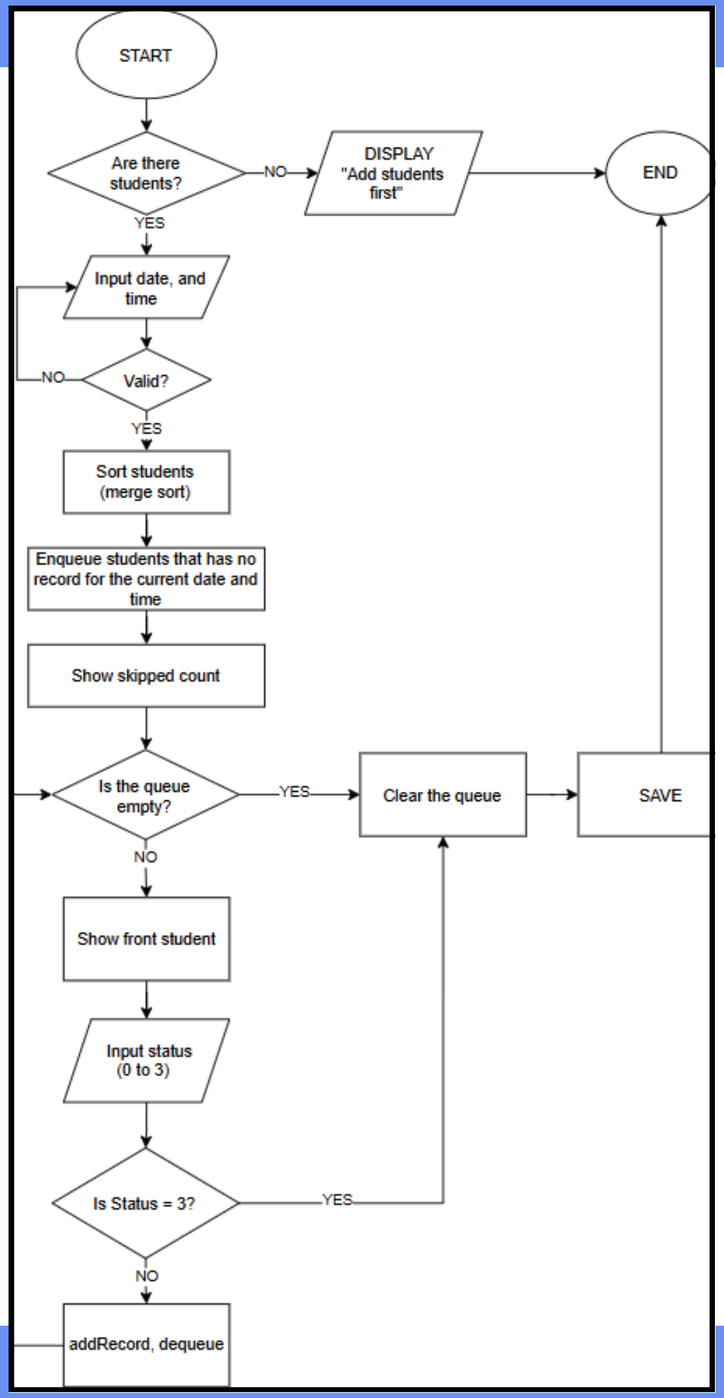
- Single Class & Single User: Handles only one class section and user at a time (no multi-account or multi-subject support).
- Console-Based: Relies on a text interface and local files (no GUI or external database).
- Manual Entry: Requires manual input by the instructor (no biometric scanning or QR code integration).
- Exact Full-Name Search: Requires exact full names for searches (case-insensitive, no partial name lookups).
- Temporary Groupings: Displays generated student groups on-screen without saving them to storage.
- Attendance-Only Focus: Excludes grading, assignment submissions, or general LMS features.

PROJECT FLOW

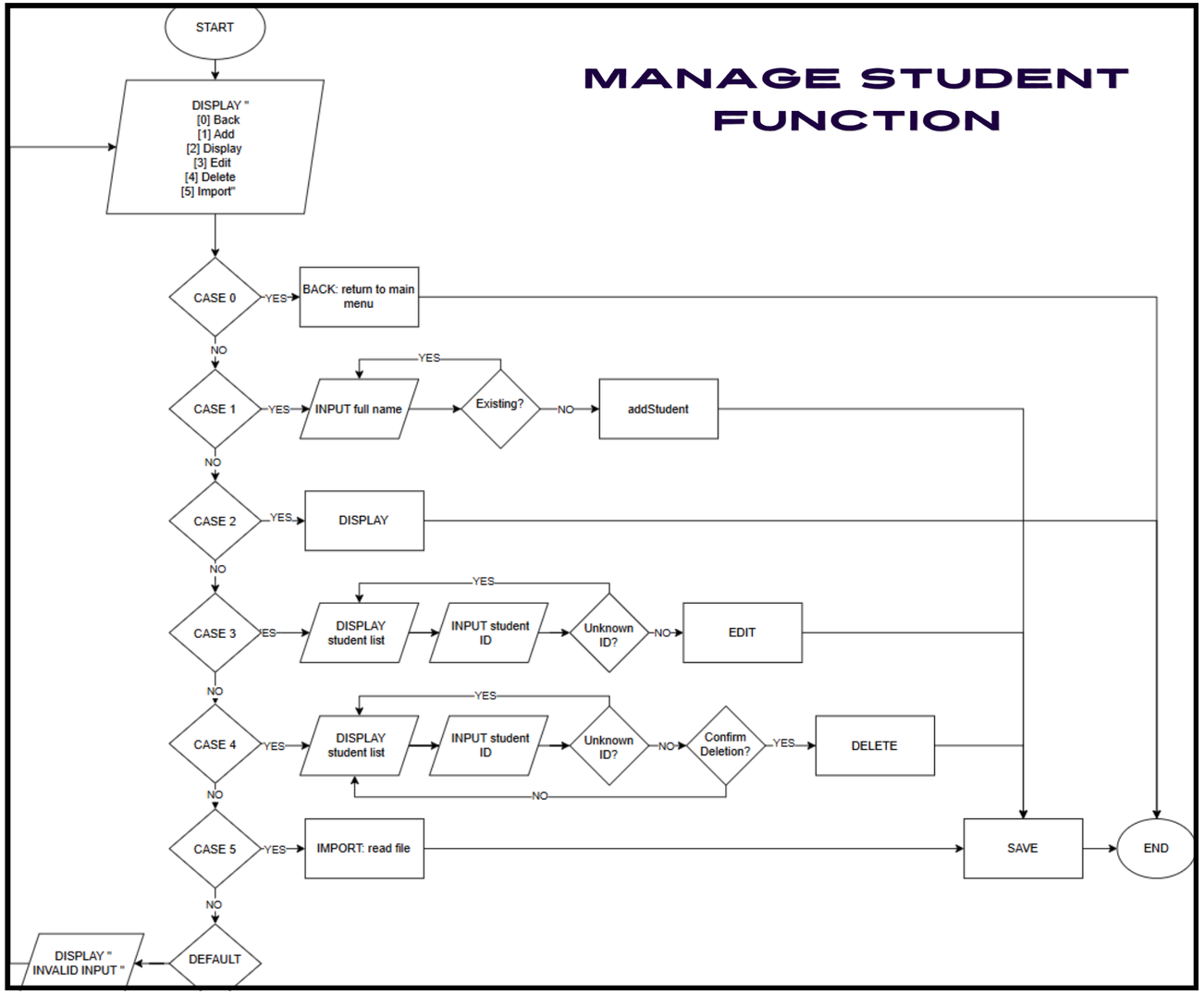
MAIN MENU

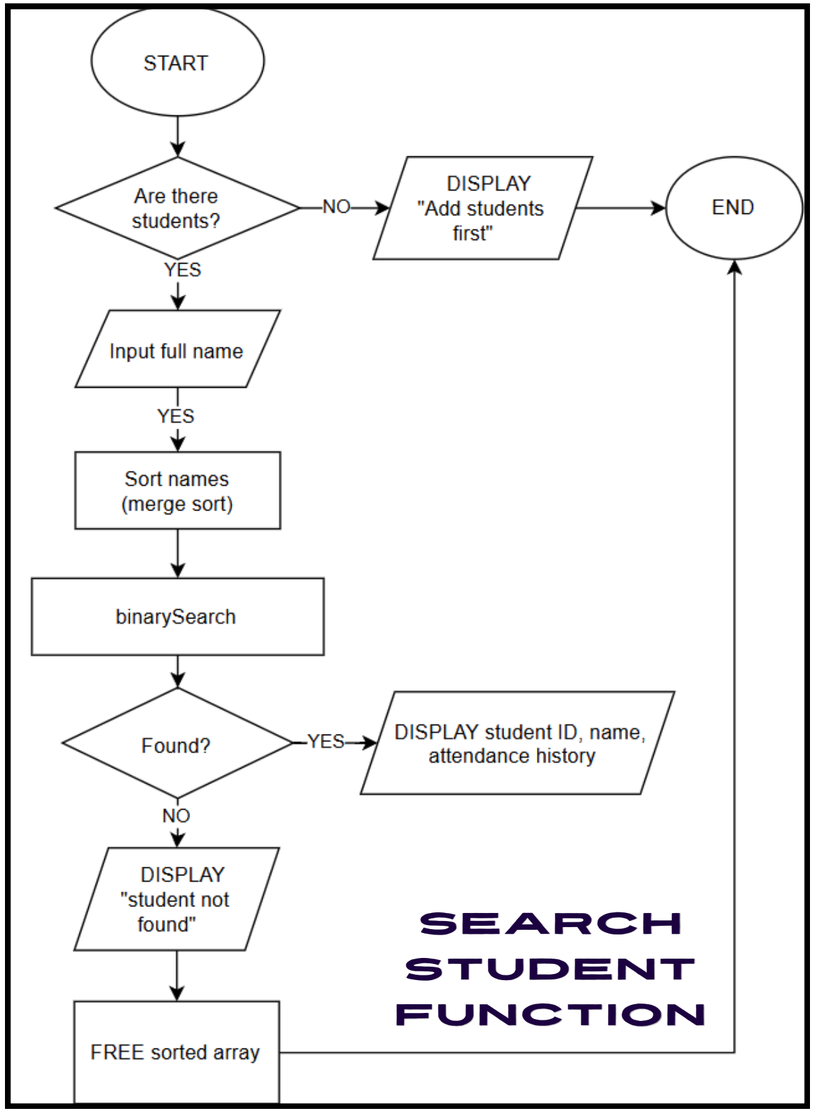


CHECK ATTENDANCE FUNCTION



PROJECT FLOW





OUTPUT DEMONSTRATION

Recomendations

- **Improve Student Searching - Using a hash table that is indexed by a unique student ID could provide an average O(1) record lookup, making the searches more efficient as the number of students increases.**

- **Strengthen Input Validation - Validate the attendance dates, status selections, as well as the student IDs before saving. Unique IDs would also distinguish students who might share the EXACT same name.**

- **Protect Attendance Records - Add confirmation prompts before deleting students or changing their already filled-in attendance entries as well as keeping backup copies of saved records so that accident changes can still be recovered.**

- **Support of Multiple Classes - Have the system expanded so that it is able to maintain separate student lists as well as attendance histories for different classes or subjects, allowing teachers to manage several different sections.**